

Audience-Aware Scientific Abstract Simplification: Metric-Guided Prompt Selection and the Limits of Self-Refinement

Drashti Bhavsar, Hard Gondaliya, Akshat Mehta, and Dev Patel

Khoury College of Computer Sciences, Northeastern University

{bhavsar.d, gondaliya.h, mehta.a, patel.dev}@northeastern.edu

Abstract

Scientific abstracts are concise records of research, but their vocabulary and compressed structure make them difficult for many non-specialists to use. We study audience-aware abstract simplification for middle-school, general-adult, and undergraduate readers. Using the SciLay corpus, we compare two fine-tuned sequence-to-sequence baselines (T5 and BART) with zero-shot Llama 3 8B Instruct, validation-based selection from four audience-specific prompts, and an iterative critique-and-revision procedure. We evaluate outputs with SARI, ROUGE-L, BLEU-4, BERTScore, Flesch–Kincaid grade-level reduction, and a source-to-output NLI entailment score. On a fixed 50-example test subset, the results show a clear trade-off. T5 obtains the strongest SARI (44.223) and entailment (0.974), but reduces grade level by only 0.98. In contrast, the zero-shot middle-school prompt reduces grade level by 7.88 while its entailment falls to 0.416. Prompt selection raises mean SARI by 0.94 and mean entailment by 0.04 across audiences, but the gains are uneven: the selected middle-school prompt becomes simpler and less supported. Self-refinement also fails to reliably recover factual support; entailment drops after the first revision for all three audiences and only partially rebounds. An exploratory audit further shows that SciLay references are 36.9% shorter than source abstracts yet are not lower in average FKGL. Together, these findings argue for multi-objective evaluation and audience-specific calibration rather than a single claim of “best” simplification.

1 Introduction

Public access to a paper does not guarantee public access to its meaning. Scientific abstracts assume familiarity with disciplinary vocabulary, compressed argument structure, statistical language, and conventions for describing uncertainty. A patient reading about a clinical intervention, a journalist preparing a report, and a student entering a new field may reach the same abstract but bring very different background knowledge. A useful simplification system therefore has to do more than replace long words. It also has to decide which ideas need unpacking, which details cannot be dropped, and

how much explanation the intended reader is likely to need.

Accuracy can be lost even when the prose becomes easier to read. A rewrite may turn an association into a causal statement, remove a qualification, leave out the study population, or present a measured improvement as evidence that an intervention “works.” Because such outputs often sound fluent and assured, these changes can be difficult to notice. Simpler wording is useful here only when the claims remain grounded in the source.

Most automatic simplification benchmarks provide one reference target, even though readers differ in what they need. Audience-aware generation instead treats the intended reader as part of the input. We consider three operational audiences: middle-school students, general adults without assumed scientific training, and undergraduates with basic scientific knowledge. The audience label is expected to affect sentence length, vocabulary, explanations, and technical detail rather than merely produce three generic paraphrases.

Large language models can follow such instructions, but their behavior depends heavily on prompt wording. Selecting a prompt after viewing test outputs risks tuning to the evaluation set. To avoid choosing prompts from test outputs, we prepare four instructions for each audience, compare them on a held-out validation sample, and carry the highest-scoring instruction into testing. We then examine a separate refinement step in which the model checks its draft against the source, identifies missing or unsupported claims, and produces a revision.

Our experiments use SciLay, a multi-journal corpus of technical and lay-oriented scientific text (Ragazzi et al., 2024). We compare T5 and BART fine-tuned on paired examples, zero-shot Llama 3 8B Instruct (Dubey et al., 2024), validation-selected Llama prompts, and self-refined outputs. We report several complementary metrics because no single score captures simplification quality, readability, semantic preservation, and factual support at once (Alva-Manchego et al., 2021; Heineman et al., 2023).

The paper makes four contributions. First, it audits the dataset and corrects the intuitive but unsupported assumption that expert lay summaries always have lower FKGL than technical abstracts. Second, it pro-

vides audience-conditioned Llama baselines and a reproducible fixed-pool prompt-selection method requiring no model training. Third, it quantifies the readability–faithfulness trade-off across audiences and supervised versus prompted systems. Fourth, it reports a negative but useful result: the implemented self-refinement loop does not reliably improve source support, showing that self-critique is not automatically a factuality safeguard.

2 Task and Research Questions

Let x denote a technical abstract, a one of three target audiences, and y the generated simplification. The task is

$$f(x, a) \rightarrow y. \quad (1)$$

A successful output should satisfy three properties. **Accessibility** means reducing avoidable lexical and syntactic burden. **Audience differentiation** means that outputs for different audiences should not collapse to one generic summary. **Source fidelity** means preserving the study’s purpose, methods, main findings, scope, and qualifications without adding unsupported claims.

These goals can conflict. Conservative systems may remain faithful but copy the source too closely. Aggressive systems may become easier to read while deleting or altering important content. We organize the study around five research questions:

1. **RQ1:** How do technical abstracts and expert lay references differ in length, long-word ratio, and FKGL?
2. **RQ2:** How do supervised sequence-to-sequence models and an instruction-tuned LLM balance reference similarity, readability, and source support?
3. **RQ3:** Does validation-based prompt selection improve over one manually chosen zero-shot prompt?
4. **RQ4:** How does the requested audience change readability gains and factual-consistency scores?
5. **RQ5:** Does iterative self-critique improve factual support after prompt selection?

3 Dataset and Exploratory Analysis

3.1 SciLay and preprocessing

SciLay contains technical and lay-oriented text from several natural-science journals (Ragazzi et al., 2024). We load the public `all` configuration. Before filtering, it contains 35,026 training examples, 4,380 validation examples, and 4,384 test examples. Records provide DOI and journal metadata together with the technical text, plain-language text, full text, topics, and keywords.

For exploratory analysis, we remove records with an empty technical or plain-text field and records in which either side has fewer than 30 whitespace-delimited tokens. This leaves 43,060 valid pairs. Journals are mapped by a hand-written rule into biology, medicine, computational science, and other. This grouping is used only for descriptive analysis; it does not affect training or prompt selection.

Domain	N	Src. FKGL	Ref. FKGL	Len. ratio
Biology	21,234	15.81	15.72	0.58
Medicine	15,732	15.44	15.99	0.66
Comp. sci.	3,657	17.36	18.79	0.74
Other	2,437	15.70	15.52	0.76
Overall	43,060	15.80	16.07	0.63

Table 1: Exploratory statistics after filtering. Length ratio is mean reference length divided by mean source length.

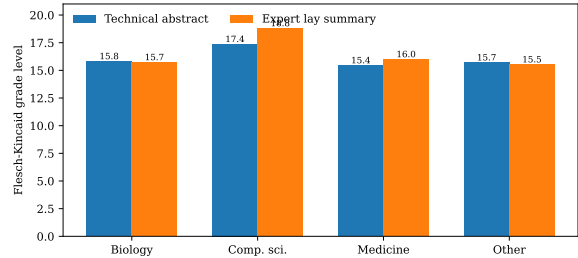


Figure 1: Mean FKGL by domain. Expert lay references are shorter, but are not uniformly lower in FKGL.

3.2 Length, vocabulary, and FKGL

Technical abstracts contain 210.44 tokens on average, while references contain 132.71. The mean reference-to-source ratio is 0.631, corresponding to a 36.9% reduction. Compression differs by domain: biology references average 58% of source length, compared with 66% for medicine, 74% for computational science, and 76% for the other group. Expert references therefore perform substantial selection and condensation rather than only replacing difficult words.

The notebook variables `src_rare` and `ref_rare` count the proportion of whitespace tokens longer than eight characters; they do not use corpus frequency. We therefore refer to this quantity as a *long-word ratio*. It decreases only slightly, from 0.242 in sources to 0.234 in references, suggesting that expert summaries often retain necessary technical terms while compressing surrounding material.

A central finding is that references are not easier according to average FKGL. The source mean is 15.80 and the reference mean is 16.07, giving a source-minus-reference difference of -0.26 . Figure 1 shows that the pattern also varies by domain. This does not imply that the references are less accessible. FKGL depends on sentence length and syllables per word (Flesch, 1948; Kincaid et al., 1975); it does not measure whether a term is explained, whether background is supplied, or whether the text is conceptually easier to follow. The RQ1 result is therefore mixed: references are much shorter and have a slightly lower long-word ratio, but a marginally higher average FKGL.

Audience	Selected framing	Val. SARI
General	Accuracy constraint	37.38
Middle school	Science teacher	40.08
Undergrad.	Teaching assistant	38.35

Table 2: Validation selection over four fixed prompts per audience ($n = 15$).

4 Methods

4.1 Fine-tuned T5 and BART

Our supervised baselines are `t5-base` (Raffel et al., 2020) and `facebook/bart-base` (Lewis et al., 2020). Both are trained on the first 5,000 training examples and evaluated during training on the first 500 validation examples. Inputs are truncated to 512 tokens and targets to 200 tokens. T5 uses the prefix `summarize:`, a learning rate of 3×10^{-4} , and batch size 8. BART uses an empty prefix, a learning rate of 3×10^{-5} , and batch size 4. Both train for at most three epochs with 300 warm-up steps, weight decay 0.01, mixed precision, gradient checkpointing, early stopping with patience 2, and four-beam decoding.

Because the training targets do not encode the three audience labels, T5 and BART are audience-agnostic lay-summary baselines. They are most directly comparable with the general-audience Llama condition rather than with a separately controlled middle-school or undergraduate system.

4.2 Zero-shot Llama 3

The instruction-tuned baseline uses `Meta-Llama-3-8B-Instruct` (Dubey et al., 2024), loaded in 4-bit NF4 quantization with double quantization and float16 computation. Generation is deterministic (`do_sample=False`) with a maximum of 256 new tokens.

We write a separate instruction for each audience. For the general audience, the instruction requests plain-language wording and assumes no prior scientific training. The middle-school version is aimed at readers aged 12–14 and emphasizes short sentences, familiar words, and brief explanations of necessary technical terms. The undergraduate version keeps useful scientific vocabulary but asks the model to unpack highly specialized language. All test evaluations use the first 50 examples of the official test split.

4.3 Metric-guided prompt selection

For each audience, we define four candidate instructions with different roles and constraints. Examples include a science-journalist framing, a classroom-teacher framing, and an explicit prohibition against adding information. Every candidate is run on the first 15 validation examples. We compute SARI against the expert reference and select the highest-scoring prompt separately for each audience.

The procedure is a fixed-pool search rather than full prompt compilation: it does not generate new instructions, demonstrations, or optimization traces. Its advantages are low cost and transparency; its main weakness is high variance from the small validation sample. The winning prompts are provided in the appendix.

4.4 Iterative self-refinement

Starting from the selected prompt output, the model receives the source abstract and a structured critique request. It is asked to identify important claims that were dropped, altered, or added without support, then revise the simplification for the same target audience. The loop allows up to three rounds.

The executed implementation has an important detail. The first revision is generated and retained without comparing its NLI score against the initial output. For later rounds, a revision is retained only when its entailment score exceeds that of the preceding accepted revision; the loop also stops if the model reports no issues or returns an empty revision. We report this behavior exactly because it affects interpretation of the refinement results.

5 Evaluation and Experimental Design

5.1 Automatic metrics

We evaluate all completed systems with six metrics. SARI (Xu et al., 2016) is the primary simplification score and evaluates n-grams kept, removed, and added relative to the source and reference. We use ROUGE-L (Lin, 2004) and BLEU-4 (Papineni et al., 2002) to quantify overlap with the expert reference. BERTScore F1 (Zhang et al., 2020) captures contextual similarity to the expert reference.

Readability change is reported as

$$\Delta\text{FKGL} = \text{FKGL}(x) - \text{FKGL}(y), \quad (2)$$

where a positive value means the generated text has a lower formula-based grade level than the source.

For source support, we use `DeBERTa-v3-base-mnli-fever-anli` (He et al., 2021; Laurer et al., 2023). The technical abstract is the premise and the generated simplification is the hypothesis; the entailment-class probability is averaged over examples. We call this measure *NLI-Entail* rather than “FactScore” because it is an automatic proxy, not a validated scientific-factuality measure. The model truncates the pair to 512 tokens and evaluates whole passages, so it can miss local contradictions, domain-specific errors, and claims removed by truncation. SummaC and AlignScore motivate the source-comparison approach but are not the exact models used here (Laban et al., 2022; Zha et al., 2023).

5.2 Comparison structure and validity safeguards

All test conditions use the same first 50 records, enabling paired descriptive comparison on a shared sample.

Prompt choice is based solely on 15 held-out validation examples; the test outputs play no role in selecting the prompt. Because the run does not preserve per-example scores or bootstrap intervals, the reported differences are descriptive; we make no claim of statistical significance or state-of-the-art performance. The findings should be read as controlled evidence on this shared sample rather than population-level estimates.

RQ2 compares T5, BART, and the general zero-shot Llama output. RQ3 compares manual and validation-selected prompts for each audience. RQ4 compares the three audiences under the same prompting strategy. RQ5 compares selected-prompt outputs with their self-refined versions and examines scores across iterations.

6 Results

6.1 Supervised models are strong but conservative

Table 3 shows that T5 obtains the strongest SARI (44.223), ROUGE-L (0.343), BLEU-4 (0.193), BERTScore (0.898), and NLI-Entail (0.974). BART follows closely on the reference-based metrics, with SARI 43.828 and BERTScore 0.895, while its NLI-Entail is lower at 0.868. Both systems are trained directly on the same source–reference relation used for evaluation, so strong overlap is expected.

Their limitation is readability control. T5 reduces FKGL by only 0.98 and BART by 0.70. These values are far below every general and middle-school Llama condition. The supervised models therefore resemble the expert references and preserve source support, but they make comparatively modest grade-level changes. Because the references themselves are much shorter than the sources, this should not be interpreted as a failure to summarize; it indicates limited control over the requested reading level.

6.2 Audience prompts expose a readability–faithfulness trade-off

The manual Llama prompts produce the expected readability ordering. The undergraduate condition has the smallest reduction (3.37), the general condition is intermediate (5.26), and the middle-school condition is largest (7.88). The model therefore responds to audience labels without additional training.

Faithfulness does not improve with simplification intensity. NLI-Entail is 0.812 for general readers, 0.616 for undergraduates, and only 0.416 for middle-school readers. The middle-school prompt asks for short sentences, simple vocabulary, and explanations, encouraging more aggressive rewriting. That behavior produces the largest accessibility gain and the weakest source support. Figure 2 visualizes the broader pattern: no system dominates all objectives. T5 lies in the high-support, low-readability-change region, while middle-school Llama outputs lie in the high-change, low-support region.

SARI is nearly flat across the three manual prompts (36.491–36.609), even though Δ FKGL varies from 3.37

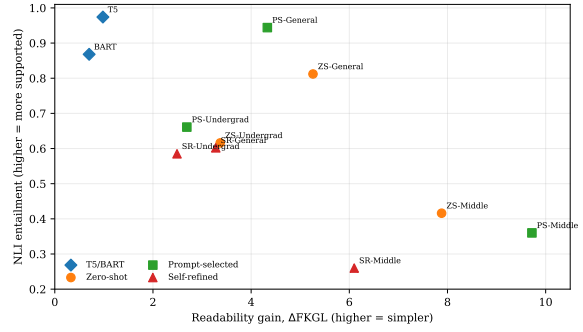


Figure 2: Readability–support trade-off across completed systems. PS denotes prompt-selected and SR denotes self-refined.

to 7.88. All audiences are evaluated against the same expert lay summary, not audience-specific references. SARI therefore has little ability to reward a rewrite simply because it is more appropriate for a child or undergraduate reader.

6.3 Prompt selection helps on average, not uniformly

Averaged across audiences, prompt selection raises SARI from 36.561 to 37.500, a gain of 0.939. Mean NLI-Entail rises from 0.615 to 0.655, while mean Δ FKGL changes only slightly from 5.50 to 5.58. These averages suggest a modest overall benefit, but the audience-level effects differ sharply.

For general readers, the selected prompt lowers SARI by 0.199 but raises ROUGE-L by 0.032, BERTScore by 0.014, and NLI-Entail by 0.132. The selected general instruction explicitly forbids unsupported additions, which may help explain the higher support score. However, the experiment does not isolate that clause, so we treat this as an interpretation rather than a causal result. Readability reduction falls from 5.26 to 4.33, making the selected output more conservative.

For middle-school readers, prompt selection raises SARI by 1.592 and increases Δ FKGL by 1.84 to the study’s maximum of 9.72. At the same time, NLI-Entail falls from 0.416 to 0.360. The SARI-selected classroom prompt therefore sharpens rather than resolves the trade-off. For undergraduates, SARI rises by 1.425 and NLI-Entail by 0.045, with a 0.68-point reduction in readability gain. This is the cleanest improvement because reference similarity and source support increase together, although simplification becomes slightly less aggressive.

RQ3 therefore has a qualified answer: fixed-pool prompt selection improves mean SARI and mean NLI support, but it is not uniformly better. Optimizing only SARI on a small validation sample can favor an output that is less source-supported. A stronger selector would use multiple objectives or maximize SARI subject to a minimum faithfulness threshold.

System	Audience	SARI	R-L	BLEU	BERT-F1	Δ FKGL	NLI-Entail
T5 fine-tuned	General	44.223	.343	.193	.898	0.98	.974
BART fine-tuned	General	43.828	.324	.178	.895	0.70	.868
Llama 3 zero-shot	General	36.582	.112	.012	.848	5.26	.812
	Middle school	36.491	.092	.005	.832	7.88	.416
	Undergraduate	36.609	.120	.016	.853	3.37	.616
Prompt-selected Llama 3	General	36.383	.144	.014	.862	4.33	.944
	Middle school	38.083	.093	.007	.828	9.72	.360
	Undergraduate	38.034	.141	.028	.861	2.69	.661
Self-refined Llama 3	General	36.912	.140	.024	.859	3.28	.602
	Middle school	38.161	.115	.019	.846	6.10	.260
	Undergraduate	37.854	.145	.031	.862	2.49	.585

Table 3: Results on the shared 50-example SciLay test subset. Bold marks the highest value in each column. A high Δ FKGL means a larger surface-readability reduction; differences are descriptive rather than significance-tested.

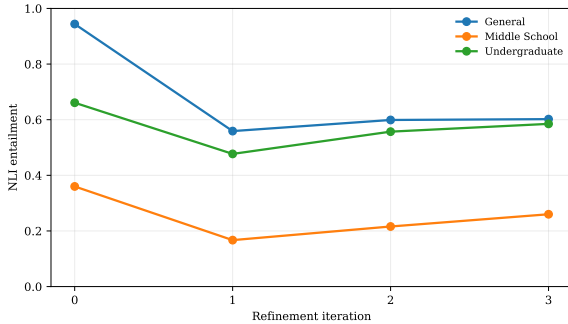


Figure 3: NLI entailment across refinement iterations. Iteration 0 is the prompt-selected output.

6.4 Self-refinement does not recover factual support

The refinement experiment produces the clearest negative result. Compared with the selected prompts, final NLI-Entail falls from 0.944 to 0.602 for general readers, from 0.360 to 0.260 for middle-school readers, and from 0.661 to 0.585 for undergraduates. SARI changes are small: general and middle-school scores rise slightly, while undergraduate SARI falls slightly.

Figure 3 shows that the largest decline occurs after the first critique-and-revision round. General entailment drops from 0.944 to 0.559, middle-school from 0.360 to 0.167, and undergraduate from 0.661 to 0.477. Later rounds partially recover but do not return to the starting score. The implementation’s unconditional first revision is an important contributor: it can replace a strong initial output before any NLI comparison is made. More broadly, the result shows that asking a model to find and repair its own errors does not guarantee that the revised text is better supported. Self-critique should be treated as a candidate-generation mechanism whose revisions require independent validation.

7 Qualitative and Behavioral Analysis

7.1 Length and audience differentiation

The metric results describe quality, but the prediction files also reveal a large difference in generation style. Table 4 reports whitespace-delimited output length. T5

Condition	Mean output words
T5 fine-tuned	43.7
BART fine-tuned	44.3
Zero-shot general	200.5
Zero-shot middle school	210.8
Zero-shot undergraduate	203.0
Prompt-selected general	139.3
Prompt-selected middle school	208.9
Prompt-selected undergraduate	192.5

Table 4: Mean output length over the shared 50-example test sample.

and BART produce compact summaries of roughly 44 words on average. The manually prompted Llama outputs are much longer, averaging 200–211 words, close to the length of the source abstracts rather than the expert references. Prompt selection changes this behavior unevenly. The selected general prompt shortens the mean output from 200.5 to 139.3 words, while the middle-school output remains close to 209 words and the undergraduate output falls modestly to 192.5.

This difference clarifies several results. The fine-tuned systems learn a highly compressed target style and obtain strong overlap scores, but their short outputs may omit explanatory context needed by a novice reader. The Llama prompts often preserve or recreate much more detail and add definitions, transitions, and analogies. Those additions can improve perceived accessibility while reducing overlap with a single concise reference. They also create more opportunities for unsupported statements. The selected general prompt is notable because its explicit accuracy constraint produces a much shorter and more source-supported output than the manual general prompt.

The audience instructions change more than FKGL. Middle-school outputs frequently use direct address (for example, “Today, we are going to talk about...”) and classroom-style analogies. Undergraduate outputs retain terminology and use a more formal explanatory style. These differences support the claim that the model responds to audience instructions, but they also expose a limitation: surface cues such as direct address do

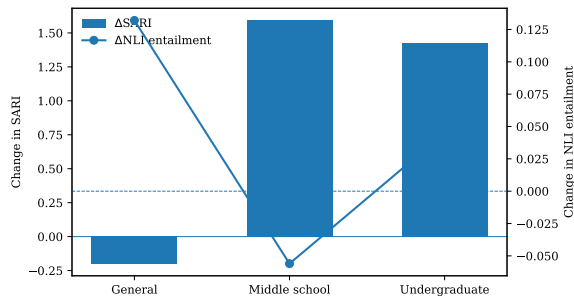


Figure 4: Change from the manual zero-shot prompt to the selected prompt. SARI and NLI support can move in opposite directions.

not demonstrate that actual readers understand the text better. A human evaluation should separate stylistic audience markers from genuine comprehension.

7.2 Representative failure modes

We manually inspected predictions to understand why aggressive simplification and refinement can lose source support. Table 5 summarizes three recurring patterns. The examples are not intended as a formal error rate; they illustrate behaviors that are visible in the generated files and help interpret the aggregate scores.

A low document-level NLI score does not always indicate an obvious scientific mistake. In one example, a middle-school rewrite explains neutrophil movement through capillaries by comparing it with cars in traffic. The comparison may aid comprehension, yet it introduces a scenario that the source never states and can therefore lower entailment. The opposite failure also occurs: a smooth rewrite can keep a fairly high score while dropping an important qualification. We therefore interpret NLI-Entail as a screening signal, not a final factuality verdict.

The qualitative patterns also differ by audience. Middle-school instructions encourage conversational metaphors, which increases the risk of invented causal explanations. The selected general prompt is shorter and explicitly limits additions. Undergraduate prompts preserve more source terminology, reducing the need for analogies but sometimes leaving too much complexity. These patterns are consistent with the quantitative trade-off and reinforce the need for audience-specific evaluation criteria.

7.3 Prompt selection and refinement as model selection

Figure 4 shows that prompt selection can move SARI and NLI in different directions. The selected general prompt sacrifices a small amount of SARI for a large NLI gain; the middle-school prompt gains SARI while losing NLI; and the undergraduate prompt improves both. A single validation objective therefore hides materially different system behaviors.

Self-refinement creates a similar selection problem. A critique may identify real omissions, but the resulting

revision is only one new candidate. Replacing the initial output automatically assumes that revision is better. The results suggest a safer design: retain the initial output, generate one or more revisions, score all candidates with independent checks, and select a Pareto-efficient option rather than the latest text. This reframes refinement as controlled candidate selection instead of an irreversible editing chain.

8 Discussion

The experiments reveal three distinct behaviors. Fine-tuned models best match the reference distribution and retain source support, but they simplify conservatively. Audience prompts create substantially larger FKGL reductions, but the most aggressive middle-school rewriting is also least supported. Prompt wording can move faithfulness even when SARI barely changes. Finally, self-refinement can damage a strong initial output, particularly when revisions are accepted without comparison to the starting point.

The audience ordering in FKGL is encouraging because it shows that an instruction-tuned model can operationalize reader profiles without additional training. Yet the low middle-school entailment is a serious warning. Simplification for younger readers often requires definitions, analogies, and compression. Each operation creates opportunities to overgeneralize, add unsupported explanations, or erase uncertainty. The correct response is not to avoid audience adaptation, but to pair it with source-grounded checks and human review.

A second issue is reference mismatch. The same expert lay summary serves as the reference for all three audiences. As a result, SARI and BERTScore may penalize appropriate audience-specific wording or reward language that does not fit the intended reader. This helps explain why zero-shot SARI is almost flat while FKGL changes substantially. Demonstrating genuine audience control will require references written for each group or comprehension tests with readers from those groups.

The dataset audit also demonstrates why FKGL should be interpreted cautiously. Expert references are 36.9% shorter yet have a slightly higher mean FKGL. Scientific names with many syllables can inflate FKGL even when the surrounding explanation is clear. At the same time, short sentences can lower the formula score even after important context has been removed. We therefore interpret FKGL together with compression, explanatory quality, reader comprehension, and factual accuracy.

9 Related Work

Lay summaries of scientific work involve content selection, compression, and explanation, not only easier vocabulary. Expert biomedical summaries are available in the PLOS and eLife corpora (Goldsack et al., 2022); BioLaySumm provides a shared evaluation setting (Goldsack et al., 2024); and SciLay extends coverage across journals while supplying the source-summary pairs

Pattern	Example from generated output	Why it matters
Invented mechanism	One middle-school TENG output explains electricity generation as the movement of “tiny particles, like dust or water molecules.”	The source attributes the device to a photoelectric–electromechanical nanogenerator using solar and tidal energy and gives no particle-motion explanation. The added analogy therefore alters the mechanism.
Category overstatement	A cancer-treatment output describes reactive oxygen species as “a special kind of medicine that can help kill cancer cells.”	The source discusses ROS-involved therapies and limitations in oxygen substrates. Calling ROS a medicine collapses a mechanism and a treatment category into one familiar but potentially misleading label.
Revision boilerplate and drift	Several refined outputs begin with “Here is a revised simplified version that addresses the issues above” and then rewrite large portions of the initial text.	The boilerplate adds no scientific value, and broad rewriting can replace a well-supported initial explanation with a more fluent but less entailed version. This pattern is consistent with the sharp first-round NLI decline.

Table 5: Representative qualitative patterns observed in the prediction files. Quoted phrases are short excerpts from model outputs.

used here (Ragazzi et al., 2024).

Text simplification can involve deleting details, replacing terms, splitting or reordering sentences, and paraphrasing (Alva-Manchego et al., 2020b,a). MUSS learns from mined paraphrases without paired simplification supervision (Martin et al., 2022), whereas controllable systems expose particular rewriting operations (Maddela et al., 2021). Our audience prompts combine several of these controls around the needs of a reader rather than adjusting only length or lexical difficulty.

DSPy treats prompts and demonstrations as components that can be optimized against an evaluation metric (Khattab et al., 2023). Our search is intentionally smaller and easier to audit: all four candidates and the selected prompt are visible. Self-Refine uses model feedback to produce successive revisions (Madaan et al., 2023). In our setting, however, a critique framed around factuality can still reduce support from the source.

No single automatic metric captures every acceptable rewrite. SARI evaluates edit behavior relative to the source and reference (Xu et al., 2016); ASSET shows that multiple different simplifications may be valid (Alva-Manchego et al., 2020a); and EASSE provides standardized diagnostics (Alva-Manchego et al., 2019). Because automatic scores do not always match human judgments (Alva-Manchego et al., 2021), SALSA evaluates edits directly (Heineman et al., 2023). SummaC and AlignScore assess whether generated claims are supported by source evidence (Laban et al., 2022; Zha et al., 2023). Our whole-passage entailment probability is a simpler proxy rather than a definitive factuality check.

10 Design Recommendations

The results suggest that audience-aware simplification should be designed as a constrained decision problem rather than a single generation step. Four practical implications follow from the experiments.

10.1 Separate content selection from audience realization

The fine-tuned models and Llama systems appear to solve different parts of the task. T5 and BART learn compact content selection from the expert references, while Llama responds more strongly to audience instructions and supplies explanations and transitions. A

future pipeline could first identify the claims that must be retained, including study design, population, main result, and qualifications, and then realize those claims at a chosen reader level. This separation would make it easier to detect when a simple explanation has dropped a required claim. It would also prevent audience control from being evaluated only through overlap with a single lay reference.

10.2 Use constrained, multi-objective prompt selection

The prompt-search experiment demonstrates why one validation metric is insufficient. Selecting on SARI helps the middle-school condition according to SARI and FKGL while lowering source support. A more suitable selector could maximize SARI subject to a minimum NLI threshold, rank candidates on a weighted combination of readability and support, or retain the non-dominated points on a Pareto frontier. The exact weighting should depend on use. The acceptable balance should depend on the application: a classroom tool may tolerate some extra explanation, whereas a patient-facing medical system should impose a much stricter penalty on unsupported additions. Prompt selection should therefore be tied to an explicit risk profile rather than a universal score.

10.3 Treat refinement as candidate generation

The first self-refinement round damages entailment across all three audiences. This does not show that model feedback is useless; it shows that replacing an output merely because it has been revised is unsafe. A better loop would preserve the initial output, generate a critique, produce one or more revisions, and compare all candidates. Revision could stop when no candidate improves a predeclared objective, and the final choice could be the best supported output that still satisfies an audience-specific readability range. Claim-level comparison would be preferable to whole-document NLI because a revision may repair one statement while damaging another.

10.4 Evaluate actual readers, not only textual proxies

The final goal is comprehension, not a low formula score. A middle-school output may use shorter sentences and

RQ	Answer from the completed experiments
RQ1	References are 36.9% shorter and have a slightly lower long-word ratio, but their mean FKGL is 0.26 points higher.
RQ2	T5 and BART lead on overlap and support; Llama produces much larger readability gains.
RQ3	Prompt selection improves mean SARI and NLI, but not every audience or metric.
RQ4	Middle-school prompts simplify most aggressively and receive the lowest NLI support.
RQ5	The implemented self-refinement loop lowers support after the first revision and does not recover the starting score.

Table 6: Summary answers to the five research questions.

still fail because the explanation assumes unfamiliar background knowledge. Conversely, a text with technical names may be understandable when those names are clearly introduced. Reader studies should therefore ask participants to answer questions about the study’s purpose, method, result, and limitations. They should also measure confidence, perceived clarity, and whether readers can distinguish what the study found from what it merely hypothesized. Such evaluation would reveal whether the audience prompts produce meaningful control or only stylistic signals.

10.5 Reproducibility and artifact scope

The final run produces prediction CSV files and metric dictionaries for T5, BART, all three zero-shot audiences, all three prompt-selected audiences, and all three self-refined audiences. It also records the selected prompts, iteration-level refinement metrics, EDA statistics, and figures. These artifacts are sufficient to reproduce the tables in this paper without preserving the multi-gigabyte model checkpoints. The notebooks remain important for documenting preprocessing, generation settings, and the exact refinement logic. Because the test and validation subsets are positional rather than randomized, a confirmatory study should first freeze explicit example identifiers and seeds, then report per-example scores and confidence intervals.

11 Limitations

The test evaluation uses the first 50 examples, while prompt selection uses the first 15 validation examples. Neither sample is randomized or stratified, and the experiment does not retain per-example scores needed for confidence intervals. T5 and BART are trained on only 5,000 of 35,026 training examples and are not audience-conditioned. These comparisons should therefore be interpreted descriptively, not as statistically confirmed differences. We partly compensate with a qualitative audit of representative outputs, but this is not a substitute for a larger randomized evaluation or reader study.

The three audiences share one expert reference. BERTScore reflects similarity to the reference rather than factual truth; FKGL does not capture prior knowledge; and the NLI model was not trained specifically to verify scientific claims. Because it processes a truncated 512-token source–output pair as a whole, it may overlook local contradictions, penalize valid compres-

sion, or miss content outside the input window. Reader groups are defined operationally through prompts and have not been validated with actual middle-school, adult, or undergraduate participants.

The self-refinement implementation also limits the conclusion. The first revision is always retained, while only later revisions are compared against the preceding accepted score. The experiment therefore evaluates the implemented loop, not the strongest possible refinement strategy. A corrected version should compare every candidate against the initial output, retain the best candidate rather than the latest one, and validate revisions with claim-level or human factuality judgments.

12 Ethics Statement

A readable scientific error can spread misinformation more effectively than a difficult abstract. Patient-, student-, and public-facing tools should keep the original source visible, clearly label generated text, preserve qualifications and uncertainty, and link important claims back to supporting evidence. High-stakes deployment requires human review. Our experiments use publicly released scientific text and do not infer private attributes about readers.

13 Conclusion and Future Work

We studied audience-aware scientific abstract simplification using fine-tuned T5 and BART, zero-shot Llama 3, metric-guided prompt selection, and iterative self-refinement. T5 achieved the strongest reference similarity and NLI support, while Llama produced much larger grade-level reductions. The middle-school prompts produced the largest grade-level reductions and, at the same time, the lowest support scores, making the trade-off between readability and faithfulness especially clear. Prompt selection improved mean SARI by 0.94 and mean entailment by 0.04, but its effects were audience-dependent. Self-refinement did not restore factual consistency and often damaged strong starting outputs.

Taken together, the results support judging systems across several objectives instead of choosing them by one simplification metric. A confirmatory study should evaluate a larger randomized sample, preserve per-example scores for confidence intervals, and require prompt candidates to satisfy a faithfulness threshold. Every refinement should be compared with the initial draft and checked at the claim level. Reader studies are equally important: participants from each target group should answer comprehension questions and flag missing or unsupported statements. That evidence is needed to determine whether automatic readability gains actually make scientific information safer and more useful.

References

- Fernando Alva-Manchego, Louis Martin, Carolina Scarton, and Lucia Specia. Easse: Easier automatic sentence simplification evaluation. In *Proceedings of EMNLP-IJCNLP: System Demonstrations*, pages 49–54, 2019.
- Fernando Alva-Manchego, Louis Martin, Antoine Bordes, et al. Asset: A dataset for tuning and evaluation of sentence simplification models with multiple rewriting transformations. In *Proceedings of ACL*, pages 4668–4679, 2020a.
- Fernando Alva-Manchego, Carolina Scarton, and Lucia Specia. Data-driven sentence simplification: Survey and benchmark. *Computational Linguistics*, 46(1): 135–187, 2020b.
- Fernando Alva-Manchego, Carolina Scarton, and Lucia Specia. The (un)suitability of automatic evaluation metrics for text simplification. *Computational Linguistics*, 47(4):861–889, 2021.
- Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, et al. The llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024.
- Rudolf Flesch. A new readability yardstick. *Journal of Applied Psychology*, 32(3):221–233, 1948.
- Tomas Goldsack, Zhihao Zhang, Chenghua Lin, and Carolina Scarton. Making science simple: Corpora for the lay summarisation of scientific literature. In *Proceedings of EMNLP*, pages 10589–10604, 2022.
- Tomas Goldsack, Carolina Scarton, Matthew Shardlow, and Chenghua Lin. Overview of the biolaysumm 2024 shared task on the lay summarization of biomedical research articles. In *Proceedings of BioNLP*, pages 122–131, 2024.
- Pengcheng He, Xiaodong Liu, Jianfeng Gao, and Weizhu Chen. Deberta: Decoding-enhanced bert with disentangled attention. In *International Conference on Learning Representations*, 2021.
- David Heineman, Yao Dou, Mounica Maddela, and Wei Xu. Dancing between success and failure: Edit-level simplification evaluation using salsa. In *Proceedings of EMNLP*, pages 3466–3495, 2023.
- Omar Khattab, Arnav Singhvi, Paridhi Maheshwari, et al. Dspy: Compiling declarative language model calls into self-improving pipelines. *arXiv preprint arXiv:2310.03714*, 2023.
- J. Peter Kincaid, Robert P. Fishburne, Richard L. Rogers, and Brad S. Chissom. Derivation of new readability formulas for navy enlisted personnel. Technical report, Naval Technical Training Command, 1975.
- Philippe Laban, Tobias Schnabel, Paul N. Bennett, and Marti A. Hearst. Summac: Re-visiting nli-based models for inconsistency detection in summarization. *Transactions of the Association for Computational Linguistics*, 10:163–177, 2022.
- Moritz Laurer, Wouter van Atteveldt, Andreu Casas, and Kasper Welbers. Deberta-v3-base-mnli-feveranli: A broad-coverage natural language inference model. Hugging Face model repository, 2023.
- Mike Lewis, Yinhan Liu, Naman Goyal, et al. Bart: Denoising sequence-to-sequence pre-training for natural language generation, translation, and comprehension. In *Proceedings of ACL*, pages 7871–7880, 2020.
- Chin-Yew Lin. Rouge: A package for automatic evaluation of summaries. In *Text Summarization Branches Out*, pages 74–81, 2004.
- Aman Madaan, Niket Tandon, Prakhar Gupta, et al. Self-refine: Iterative refinement with self-feedback. In *Advances in Neural Information Processing Systems*, volume 36, 2023.
- Mounica Maddela, Fernando Alva-Manchego, and Wei Xu. Controllable text simplification with explicit paraphrasing. In *Proceedings of NAACL-HLT*, pages 3536–3553, 2021.
- Louis Martin, Angela Fan, Eric de la Clergerie, et al. Muss: Multilingual unsupervised sentence simplification by mining paraphrases. In *Proceedings of LREC*, pages 1651–1664, 2022.
- Kishore Papineni, Salim Roukos, Todd Ward, and Wei-Jing Zhu. Bleu: A method for automatic evaluation of machine translation. In *Proceedings of ACL*, pages 311–318, 2002.
- Colin Raffel, Noam Shazeer, Adam Roberts, et al. Exploring the limits of transfer learning with a unified text-to-text transformer. *Journal of Machine Learning Research*, 21(140):1–67, 2020.
- Luca Ragazzi, Paolo Italiani, Gianluca Moro, and Mattia Panni. What are you token about? differentiable perturbed top-k token selection for scientific document summarization. In *Findings of ACL*, pages 9427–9440, 2024.
- Wei Xu, Courtney Napoles, Ellie Pavlick, Quanze Chen, and Chris Callison-Burch. Optimizing statistical machine translation for text simplification. *Transactions of the Association for Computational Linguistics*, 4: 401–415, 2016.
- Yuheng Zha, Yichi Yang, Ruichen Li, and Zhiting Hu. Alignscore: Evaluating factual consistency with a unified alignment function. In *Proceedings of ACL*, pages 11328–11348, 2023.

Tianyi Zhang, Varsha Kishore, Felix Wu, Kilian Q. Weinberger, and Yoav Artzi. Bertscore: Evaluating text generation with bert. In *International Conference on Learning Representations*, 2020.

A Selected Prompts and Run Audit

The fixed-pool search selected the following instructions on the 15-example validation sample. The source abstract was inserted at {abstract}.

General adult. “Simplify this scientific text so any adult can understand it. Do not add information that is not in the original: {abstract} Simplified:”

Middle school. “You are a science teacher explaining this to a middle school class. Keep it accurate and easy: {abstract} Explanation:”

Undergraduate. “You are a teaching assistant. Rewrite this for undergraduates, preserving accuracy while improving readability: {abstract} Rewrite:”

For reproducibility, all test comparisons use the first 50 official test examples. Prompt selection uses the first 15 validation examples. Llama generation is greedy with a 256-token output cap and 4-bit NF4 quantization. T5 and BART use the first 5,000 training and 500 validation examples, at most three epochs, 512-token inputs, 200-token targets, and four-beam decoding. Self-refinement permits three critique-and-revision rounds.

Component	Evidence in final run
Dataset EDA	Completed for 43,060 filtered pairs
T5	Fine-tuning, predictions, and metrics completed
BART	Fine-tuning, predictions, and metrics completed
Llama zero-shot	Completed for three audiences
Prompt search	Four candidates per audience; completed
Self-refinement	Three audiences and three rounds completed

Table 7: Audit of the completed experimental artifacts.